7 (2026) 100087

Contents lists available at ScienceDirect

Geodata and AI

journal homepage: www.elsevier.com/locate/geoai

The role of geotechnics researchers in the era of rapid AI advancement

Stephen Wu [a,b,*], Iason Papaioannou [c,*], Yelu Zhou [c,d], Yanzhe Liu [d], Mingliang Zhou [d], Dongming Zhang [d], Hongwei Huang [d]

[a] Research Organization of Information and Systems, The Institute of Statistical Mathematics, 10-3 Midori-cho, Tachikawa, Tokyo, 190-8562, Japan
[b] The Graduate University for Advanced Studies (SOKENDAI), 10-3 Midori-cho, Tachikawa, Tokyo, 190-8562, Japan
[c] Engineering Risk Analysis Group, Technical University of Munich, Arcisstrasse 21, Munich, 80290, Germany
[d] College of Civil Engineering, Tongji University, 1239 Siping Road, Shanghai, 200092, China

ARTICLE INFO

Keywords:
Transfer learning
Probabilistic machine learning
Data privacy
PINNs
LLMs

ABSTRACT

The rapid advancement of artificial intelligence (AI) presents both significant opportunities and critical challenges for geotechnical engineering research. This paper explores the evolving role and responsibilities of geotechnics researchers in the AI era, highlighting how AI can substantially enhance information acquisition during the investigation, design, construction, and operation phases of geotechnical projects. It identifies key AI technologies — such as transfer learning, probabilistic machine learning, data privacy techniques, physics-informed neural networks (PINNs), and large language models (LLMs) — that are particularly valuable in addressing complex, uncertain, and data-scarce geotechnical scenarios. Furthermore, the paper emphasizes essential guidelines for effectively integrating these AI methods into practice, including the importance of rigorous validation, interdisciplinary collaboration, and establishing standardized benchmarks. Ultimately, strategic fusion of AI capabilities with deep geotechnical domain expertise promises to significantly advance the field, enabling more reliable, efficient, and interpretable engineering solutions.

1. Introduction

Geotechnical engineering is an inherently interdisciplinary research field that requires researchers to make use of knowledge from geoscience to solve practical engineering problems with complex boundary conditions. A typical scientific approach aims to develop rigorous theories to explain the observed physical world. In reality, the physical world is often too complex to model directly; thus, most scientific theories begin from simplified problem setups. For instance, theories in soil mechanics often start by assuming homogeneous and isotropic soil conditions, whereas in real-world scenarios, soils exhibit highly heterogeneous and anisotropic characteristics, with varying compositions and uncertain spatial distributions (Baecher & Christian, 2008; Phoon & Kulhawy, 1999). Similarly, traditional slope stability analyses commonly assume simplified slip surfaces, whereas actual landslides involve intricate and irregular failure surfaces influenced by environmental factors such as rainfall and vegetation (Griffiths et al., 2009). On the contrary, engineers aim to provide solutions to real-world problems; hence, they need to make decisions under highly uncertain information and imperfect theoretical models. One of the most important roles of researchers in geotechnical engineering is to bridge the gap between rigorous scientific approaches and the practical demand to develop solutions for complex real-world problems (Phoon et al., 2022). For example, researchers focus on refining numerical models for slope stability analysis by addressing the critical tradeoff between model complexity and computational efficiency, particularly by incorporating realistic spatial variability of soil properties and dynamic environmental factors such as rainfall patterns (Lei et al., 2023). Successfully navigating this tradeoff significantly enhances the accuracy of landslide risk assessments and informs effective mitigation strategies. However, there are rising challenges in fulfilling this critical role amid the rapid development of artificial intelligence (AI) technologies.

The rapid advancement of AI technologies presents several significant research challenges for geotechnical researchers. Firstly, the strong scaling behavior of black-box deep neural networks has led to a growing reliance on performance-driven AI approaches, where researchers often assume that increasing the training data alone will sufficiently improve model predictions, despite lacking thorough comprehension of the underlying mechanisms. Secondly, the allure of big data combined with deep learning creates a misleading perception that accumulating large datasets can inherently solve complex problems, potentially diverting researchers from critically evaluating the purpose and relevance of data

* Corresponding authors.
E-mail addresses: stewu@ism.ac.jp (S. Wu), iason.papaioannou@tum.de (I. Papaioannou), yelu.zhou@tum.de (Y. Zhou), 2151754@tongji.edu.cn (Y. Liu), zhoum@tongji.edu.cn (M. Zhou), 09zhang@tongji.edu.cn (D. Zhang), huanghw@tongji.edu.cn (H. Huang).

https://doi.org/10.1016/j.geoai.2026.100087
Received 16 September 2025; Received in revised form 28 February 2026; Accepted 17 April 2026
Available online 24 April 2026
3050-483X/© 2026 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

collection. Thirdly, the sheer volume of automated publications and new AI tools emerging in rapid succession increases the overhead needed to effectively compress information and extract meaningful insights. Finally, the accelerating pace of machine learning innovation overwhelms geotechnical researchers, often leaving little time for deep understanding or meaningful modification of emerging methodologies, as they are frequently occupied with merely testing and adopting the latest techniques.

This paper aims to address the role of geotechnical researchers amid rapid AI advancements by examining how AI technologies can enhance information acquisition and providing practical guidelines for conducting AI application research in geotechnical engineering. Section 2 discusses AI-enhanced information acquisition, highlighting critical geotechnical applications where expert knowledge is indispensable. It covers intelligent data acquisition during the investigation and design phases, such as intelligent identification of geological conditions and geotechnical data acquisition technologies, and real-time monitoring and optimization during construction and operation phases, including structural health monitoring and quality control. Section 3 presents challenges and guidelines for AI research within geotechnics, offering insights from the authors' experiences and discussing topics such as data privacy, transfer learning, metrics for probabilistic modeling, convergence of physics-informed neural networks (PINNs), and the use of large language models (LLMs). Section 4 provides a broader discussion, advocating for community-driven initiatives, interdisciplinary rather than multidisciplinary research, and collective efforts toward meaningful benchmark data development. Finally, Section 5 summarizes the key conclusions drawn from the discussions and recommendations presented.

2. On-going trends in geotechnics AI research

The distinctive characteristics of geotechnical engineering – including inherent spatial variability of geomaterials, sparse and costly data acquisition, partial observability of subsurface conditions, and safety-critical decision-making under epistemic uncertainty – create methodological demands that differ from many other engineering disciplines. Unlike systems composed of manufactured and repeatable materials, geotechnical systems are site-specific and often irreproducible, with limited opportunities for controlled experimentation. These features make uncertainty quantification, data efficiency, physics consistency, and real-time information assimilation particularly critical. In this context, AI technologies do not merely serve as performance-enhancing tools, but as enabling frameworks for managing variability, integrating heterogeneous information sources, and supporting decision-making under uncertainty. The following sections illustrate how recent AI developments are increasingly aligned with these distinctive needs of geotechnical practice.

This section discusses recent advancements in AI technologies across two critical phases of geotechnical engineering–investigation and design, and construction and operation. It highlights the growing impact of AI methods on enhancing geological condition identification, geotechnical data acquisition, real-time monitoring, process optimization, quality control, and structural health monitoring.

2.1. Investigation and design phase

The geotechnical investigation and design phase involves stratum identification and geotechnical parameter determination. With the integration of AI technologies, both the efficiency and accuracy of geological data acquisition have been significantly improved. More details are discussed in the following two subsections and summarized in Fig. 1.

2.1.1. Intelligent identification of geological conditions

Recent advances in AI have enhanced geological data interpretation, significantly improving stratigraphic classification and spatial modeling accuracy. Because of different construction and cost constraints, traditional geological stratum identification primarily relies on limited borehole drilling data, resulting in insufficient spatial coverage of geological conditions (Jiang et al., 2024). Two research directions have emerged to address this limitation: (1) enhancing borehole data interpretation using AI techniques, and (2) integrating indirect exploration methods, such as remote sensing and geophysical prospecting, to expand the spatial scope of subsurface investigation.

AI-enhanced approaches including the incorporation of deep neural networks and ensemble frameworks, have been widely applied to stratigraphic classification and spatial modeling, improving automation and processing efficiency. Machine learning models trained on Cone Penetration Tests (CPT), electrical resistivity, and seismic data can extract spatiotemporal features that help identify stratigraphic boundaries and their spatial distributions (Zhang et al., 2020; Zinas et al., 2025). The incorporation of Markov models, graph-based representations, and uncertainty quantification further supports three-dimensional stratigraphic reconstruction with improved geological consistency (Zhang et al., 2021a). In addition, integrating indirect sensing technologies such as remote sensing imagery, seismic surveys, ground-penetrating radar (GPR), and transient electromagnetic methods with AI-driven automation has expanded spatial coverage and accuracy in subsurface geological investigations (Nie et al., 2024; Yeh et al., 2024). AI models integrating multi-source geological data support coherent and reliable three-dimensional stratigraphic reconstructions even under sparse data conditions.

2.1.2. Intelligent geotechnical data acquisition technologies

AI-driven methods have significantly advanced the quantitative characterization of geotechnical parameters obtained through in-situ and laboratory tests. Aiming to obtain geotechnical parameters related to strength, deformation characteristics, and mechanical response, traditional geotechnical field tests (Cheng et al., 2024), including Standard Penetration Tests (SPT), CPT, and vane shear tests, alongside laboratory experiments like triaxial shear tests, have benefited from data-driven modeling approaches such as neural networks, support vector machines, and tree-based regressors (Zhang et al., 2021b). PINNs, incorporating physical constraints within their learning architecture, particularly enhance reliability and consistency with geomechanical behavior (Zhang et al., 2023c). AI-enhanced inversion workflows now effectively account for spatial variability and improve parameter estimates even in challenging geological contexts.

2.2. Construction and operation phase

The integration of sensor networks and AI techniques has enabled intelligent monitoring, real-time risk detection, and data-driven decision support. Current methods can be categorized into three areas: real-time construction monitoring, construction process optimization and quality control, and long-term structural health evaluation. More details are given in the following three subsections and summarized in Fig. 2.

2.2.1. Real-time monitoring data during construction

AI technologies have improved real-time monitoring capabilities during geotechnical construction phases through the deployment of sophisticated sensor networks, including fiber optic sensors, MEMS tilt sensors, and laser displacement sensors integrated with wireless sensor networks (WSN) and Internet of Things (IoT) platforms (Nie et al., 2023). Dynamic sensing techniques, such as acoustic emission and vibration monitoring, further improve detection of mechanical responses from surrounding rock and tunnel linings (Zhang et al., 2024b). AI-based anomaly detection models, including autoencoders and other deep neural network architectures, analyze multivariate monitoring data to identify abnormal patterns and provide early warnings for risks like face instability and segmental deformation (Xia et al., 2025). Local data pro-

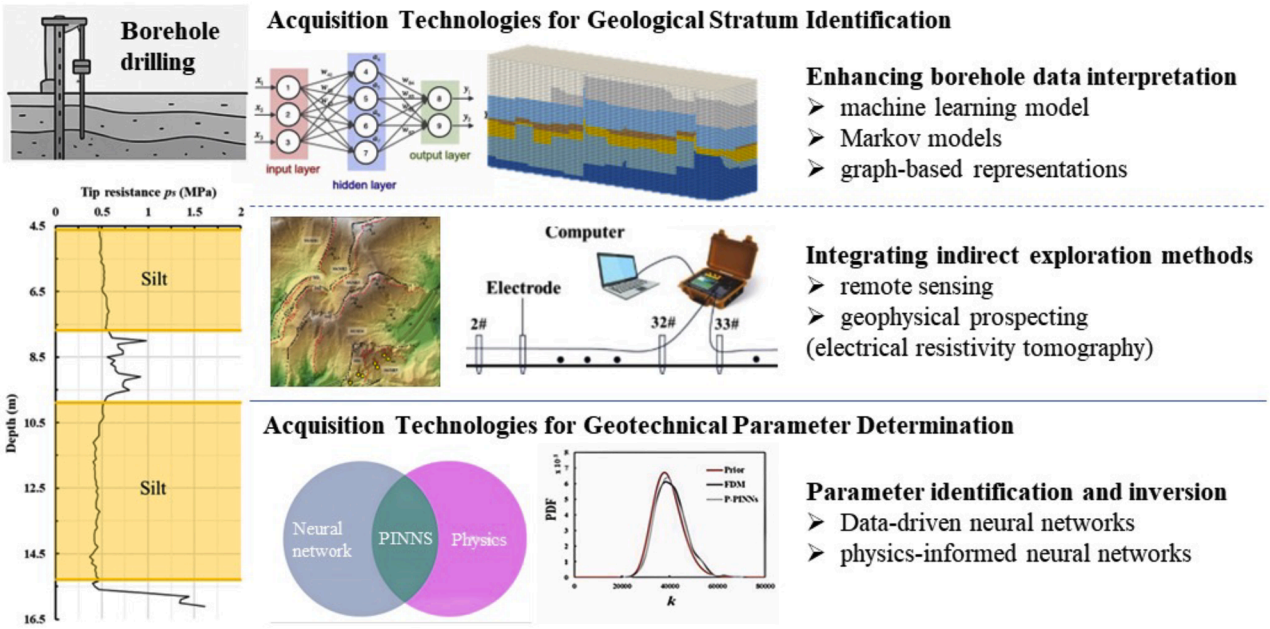


Fig. 1. Overview of acquisition technologies for the investigation and design phase, covering geological stratum identification and geotechnical parameter determination, and integrating borehole data, indirect exploration, and data-driven inversion methods.

cessing facilitated by edge computing further enhances real-time responsiveness and bandwidth efficiency.

2.2.2. *Intelligent optimization of construction process and quality control*

Geotechnical construction processes and quality control have been significantly optimized through AI-integrated data acquisition and analytical approaches. Real-time operational parameters like thrust force, torque, advance rate, and alignment angles are monitored and dynamically adjusted through predictive models based on PINNs (Chang et al., 2024). Deep learning-based image recognition techniques efficiently detect construction defects, misalignment, and quality deviations (Huang et al., 2024; Zhang et al., 2023a). Moreover, integrated AI-BIM platforms and intelligent machinery facilitate standardized quality control and process optimization, significantly enhancing construction management efficiency.

2.2.3. *Structural health monitoring on lifetime performance*

Long-term structural health monitoring in geotechnical engineering now extensively employs comprehensive, multi-modal sensor networks, including strain gauges, accelerometers, temperature sensors, image-based systems, and 3D point cloud scanning technologies (Zhang et al., 2023b). AI-driven methodologies such as Vision Transformer (ViT) models (Dosovitskiy et al., 2021) significantly enhance defect detection and quantification accuracy for structural elements (Huang et al., 2025). Combined with time-series analysis, these techniques support differentiation between transient disturbances and long-term deterioration. Probabilistic deep learning models are applied to quantify uncertainty in safety assessment and service life prediction, while clustering algorithms such as K-means++ enable unsupervised classification of structural health states and identification of critical thresholds (Zhou et al., 2023). Blockchain technology further ensures secure, traceable, and reliable data governance for structural monitoring records (Zhang et al., 2024a).

3. Challenges and guidelines of AI research in geotechnics

In the context of rapidly evolving AI technologies, geotechnical researchers face several pivotal challenges and opportunities. Effectively integrating AI into geotechnical practice requires careful attention to data quality, accessibility, and privacy, alongside robust methods to ensure that AI models generalize appropriately to diverse geotechnical conditions. This section discusses these critical aspects in detail, offering guidelines and highlighting the necessity of rigorous evaluation methods. The aim is to highlight the evolving role and responsibilities of geotechnical researchers in effectively navigating and leveraging rapid AI advancements within the field, thereby providing non-trivial value to geotechnical engineering that practical engineers alone may not readily achieve.

3.1. *Data quality and accessibility challenges*

Handling the complexity and uncertainty of geotechnical engineering data is a fundamental challenge for AI applications. Small sample and imbalanced data learning methods require specialized algorithms developed for geotechnical engineering characteristics. Transfer learning offers a solution to apply deep learning with limited data, and probabilistic machine learning provides uncertainty quantification capability when solving geotechnical problems. In some scenarios, data sparcity is caused by lack of data sharing due to data privacy or security concerns. Therefore, there is an urgent need to develop privacy protection and data security strategies to promote AI applications in geotechnical engineering. More details on these topics are discussed in the following three subsections.

3.1.1. *Transfer learning*

Transfer learning is a machine learning approach where a model trained on one problem is repurposed for a different but related task. In practical terms, it involves fine-tuning a pre-trained model (often one trained on large datasets from another domain) to solve a specific geotechnical problem (often with limited data). This strategy has been eagerly adopted in geotechnical engineering to tackle data scarcity, with studies applying transfer learning to tasks like soil property prediction (Zhang et al., 2022) and slope stability analysis (Azmoon et al., 2021; Zhang et al., 2024c), thereby reducing the need for large domain-specific datasets. However, many such studies adopt transfer learning without thoroughly validating that the pre-trained model's knowledge is applicable to their target geotechnical context. This can be problematic if key differences between the source and target domains are

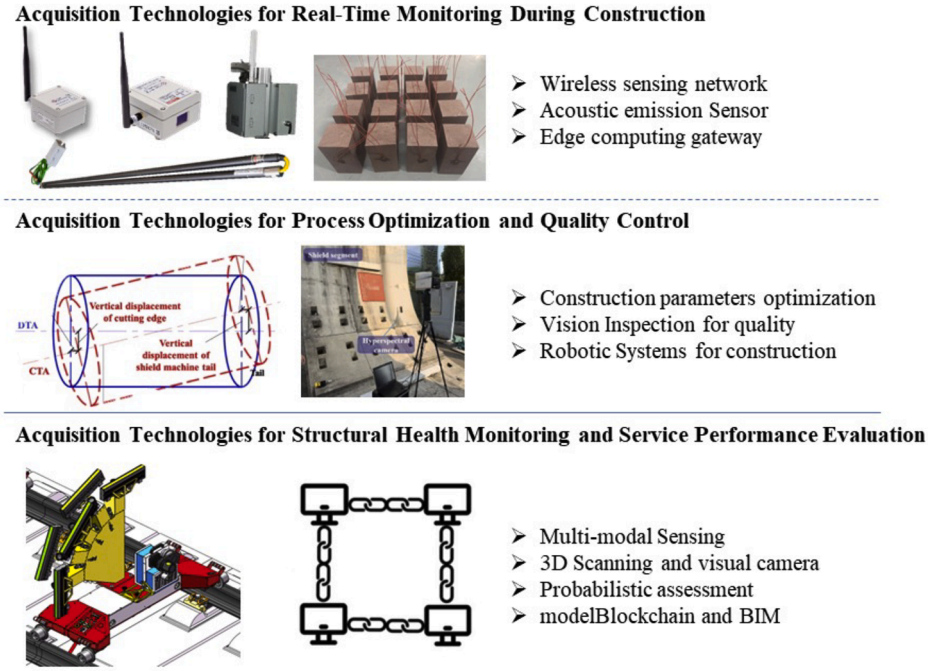


Fig. 2. Overview of acquisition technologies for the construction and operation phase, covering real-time sensing, process optimization, and structural health monitoring.

overlooked, potentially leading to suboptimal or even misleading results. Beyond empirical performance comparison, geotechnical domain knowledge should actively inform transfer learning design — for example, by selecting source domains with comparable geological settings, stress regimes, or constitutive behavior, and by constructing physics-informed feature representations. Importantly, such domain-informed assumptions must themselves be quantitatively validated, rather than implicitly trusted, to ensure that transferred knowledge improves generalization rather than introducing hidden bias.

For robust use of transfer learning in geotechnics, two critical considerations must be addressed. Scaling laws are the first: these empirical relationships help quantify how model performance improves as training data scale up. Recent research shows that as the size of a pretraining dataset increases, the prediction error on target tasks often decreases following a power-law trend (Mikami et al., 2022). In other words, expanding the source data yields progressively smaller improvements, but such scaling behavior can be analyzed to estimate generalization gains and to determine how much data might be needed for a desired accuracy. Equally important is guarding against catastrophic forgetting (Kirkpatrick et al., 2017; McCloskey & Cohen, 1989) and negative transfer (Wang et al., 2019) when fine-tuning models. Catastrophic forgetting occurs when a model "forgets" previously learned knowledge and overfits to the new data during transfer, eroding the benefits of pre-training. This is often a serious issue in the context of continual learning (De Lange et al., 2022). Negative transfer refers to the situation where knowledge from an insufficiently related source domain actually impairs performance on the target task. If the source and target domains are poorly matched (for example, using a generic image model for a very specialized geotechnical imagery task with unique image features), the transferred model might perform worse than a model trained from scratch. These risks underscore the need for careful validation: researchers should ensure the source domain is sufficiently relevant and consider strategies to mitigate forgetting, so that transfer learning genuinely aids geotechnical predictions.

To concretely illustrate the application of scaling laws, a demonstrative study was conducted utilizing the YOLOv8 object detection framework. The study aimed to quantify how pretraining dataset size impacts the performance of transfer learning in detecting tunnel leakages from digital and line-scan camera imagery (Fig. 3). Two distinct datasets were employed: a larger pretraining dataset consisting of approximately 3700 tunnel lining images collected by the team at Tongji University, and a smaller open-source finetuning dataset comprising 125 images. Both datasets were systematically split into training, validation, and test sets with an 8:1:1 ratio, ensuring methodological rigor. The experiment varied the pretraining dataset size across four levels–10, 100, 1,000, and 3000 images–and each level underwent repeated trials (10 repetitions) to ensure statistical reliability. Evaluation metrics, specifically the F1 and Average Precision (AP) scores, demonstrated a clear and substantial improvement in model performance as the pretraining dataset size increased (Fig. 4), highlighting the predictive utility and practical relevance of scaling laws in transfer learning for geotechnical engineering applications.

3.1.2. Metric for probabilistic predictive modeling

Predictive machine learning models are increasingly employed to guide decisions related to site characterization and structural safety. In geotechnical engineering, this requirement is further shaped by established reliability-based design philosophies, where safety is evaluated through failure probability and reliability indices. Consequently, probabilistic learning frameworks should be aligned with domain-specific risk metrics rather than relying solely on generic prediction accuracy. In such contexts, it is essential to not only make accurate predictions but also to quantify the uncertainty associated with them. Probabilistic machine learning addresses this need by modeling the full distribution of possible outcomes through accounting for uncertainties in data, models and learning methods, thus enabling more informed and robust decisions than deterministic approaches (Gressmann et al., 2018). Moreover, probabilistic machine learning approaches are often inherently able to deal with the multivariate, sparse and incomplete nature of geotechnical data.

In supervised learning, the goal is to estimate parameters θ of a mapping $g : \mathcal{X} \rightarrow \mathcal{Y}$, where $\boldsymbol{X} \in \mathcal{X}$ denotes independent variables (features) and $Y \in \mathcal{Y}$ is the dependent (target) variable. The model is trained on a dataset consisting of paired observations of inputs and corresponding

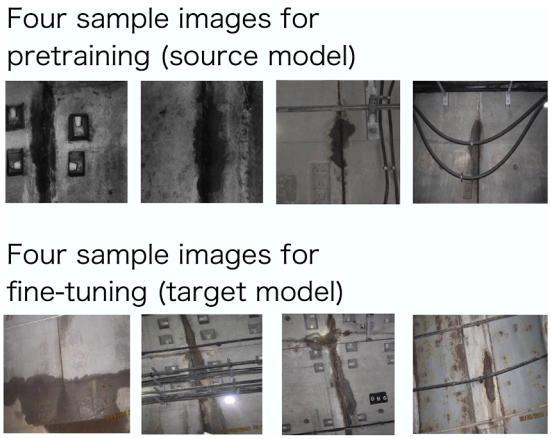


Fig. 3. Sample images of the pretraining and fine-tuning data.

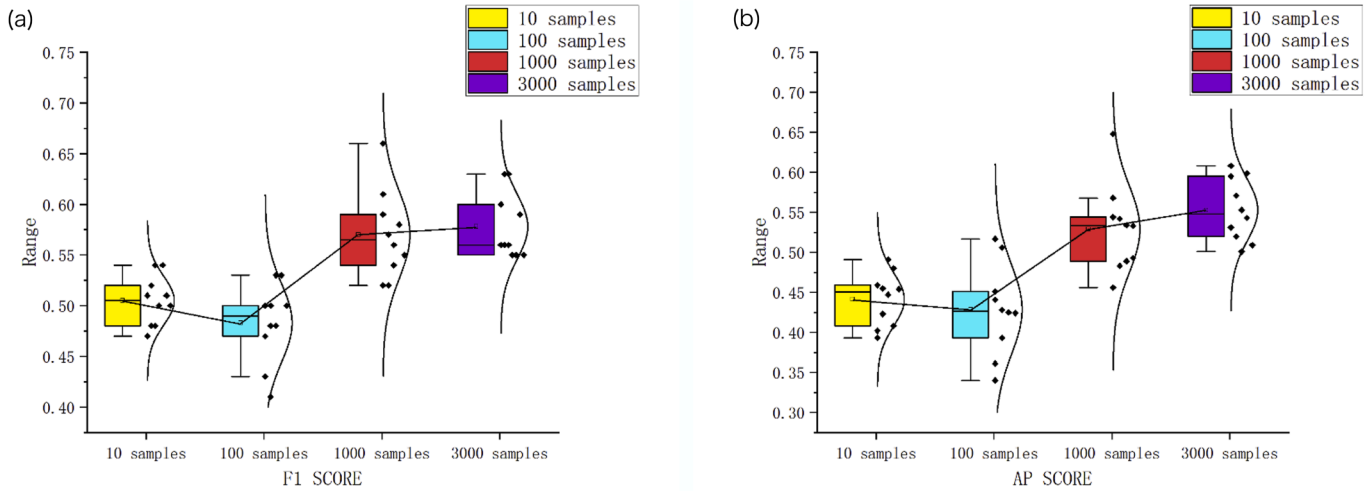


Fig. 4. Performance of the fine-tuned models on test data as a function of number of pretraining data used, where the performance is evaluated based on (a) F1 score and (b) AP score.

outputs. Deterministic models return a point prediction $\hat{y} = g(\boldsymbol{x}|\boldsymbol{\theta})$ for a new input $\boldsymbol{x}$, without accounting for uncertainty. In contrast, probabilistic models learn a mapping $g : \mathcal{X} \to \text{Distr}(\mathcal{Y})$ that outputs a conditional probability density function (PDF) $f(y|\boldsymbol{x}, \boldsymbol{\theta})$. This PDF captures the full distribution of the dependent variable conditioned on $\boldsymbol{x}$, from which point estimates such as the mean or mode can still be derived. By explicitly modeling prediction uncertainty, probabilistic approaches are better suited for safety-critical applications.

Model performance is typically assessed by evaluating predictions on unseen data using a loss function $L : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_{\geq 0}$, comparing predicted and observed values. The expected generalization loss is given by:

$$\epsilon_L = \mathbb{E}[L(g(\boldsymbol{X}), Y)], \tag{1}$$

where the expectation is estimated based on a test dataset. In deterministic settings, a common choice is the squared loss, $L(\hat{y}, y) = (\hat{y} - y)^2$, whose average yields the mean squared error (MSE). However, squared loss does not reflect prediction confidence. In probabilistic models, this limitation is addressed using probabilistic scoring rules (Murphy, 2012). The most common loss function in this context is the log-loss (or cross-entropy loss), defined as:

$$L(f(\cdot|\boldsymbol{x}), y) = -\ln f(y|\boldsymbol{x}). \tag{2}$$

The log-loss penalizes low predictive probability at the observed value y: it increases when the model assigns low density to the true outcome or produces overly uncertain (flat) distributions. Consequently, the expected log-loss captures both the central tendency and uncertainty of the predictive distribution, making it a more appropriate measure for evaluating probabilistic models. In mathematical terms, the log-loss is a proper scoring rule, which implies that the expected score is uniquely maximized by the true data-generating distribution (Gressmann et al., 2018).

To demonstrate the effect of the choice of loss function on model evaluation, a 5-fold cross-validation study was conducted using three probabilistic regression methods: Gaussian Process Regression (GPR), Bayesian Linear Regression (BLR), and an ensemble of regression trees. The methods were applied on the F-CLAY/7/216 dataset in TC304dB, transformed according to D'Ignazio et al. (2016), to predict the mobilized undrained shear strength as a function of 6 independent variables. Fig. 5 reports the average MSE and log-loss as well as box plots of the variability across folds. Similar average MSE values were observed for GPR and BLR across folds, with GPR displaying slightly lower error but also greater variability, as shown in the box plots. In contrast, a clearer separation emerged when comparing log-loss: BLR achieved a noticeably lower average log-loss compared to both BLR and ensemble trees. This result illustrates how models with comparable point prediction accuracy can differ in their ability to represent predictive uncertainty–an aspect revealed only through the use of a proper metric such as log-loss.

3.1.3. *Data privacy*

Data privacy has become increasingly critical in geotechnical machine learning research, driven by the sensitive nature of the data

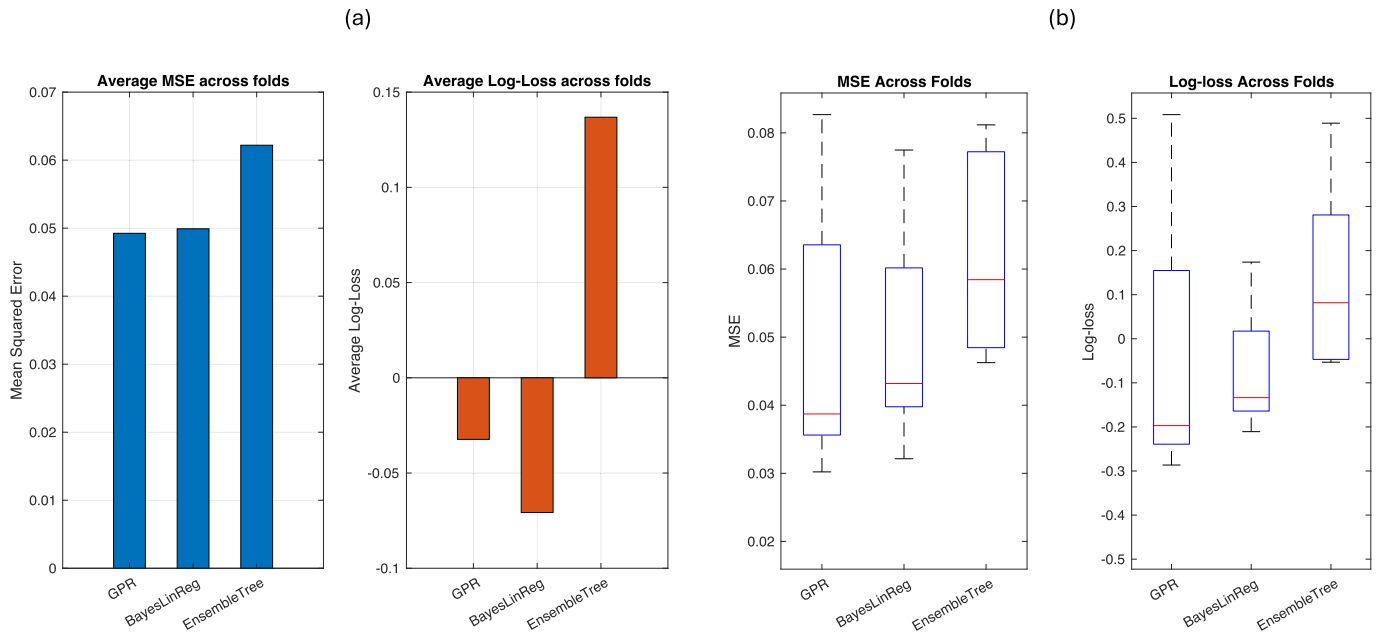


Fig. 5. Comparison of mean square error and mean log-loss obtained through 5-fold cross validation with different probabilistic predictive machine learning methods. (a) average error metrics (b) box plots of error metrics across folds.

involved–including proprietary site investigation reports, detailed subsurface profiles, and infrastructure health monitoring datasets. Exposing such sensitive data could result in significant financial, competitive, or even security risks. Despite these concerns, the adoption of data privacy methods within the geotechnical community has significantly lagged behind sectors like healthcare and finance (Murakami et al., 2025). This slow adoption arises primarily due to limited awareness and understanding of privacy-preserving technologies among geotechnical engineers, along with an absence of established standards or guidelines. Consequently, the lack of robust privacy protection frameworks discourages data sharing and collaboration, creating significant barriers for adopting advanced machine learning methods. To overcome this challenge, it is essential to systematically investigate and promote data privacy techniques tailored to geotechnical engineering, facilitating secure collaboration and maximizing the value derived from shared datasets.

Key privacy-preserving technologies include Differential Privacy (DP) (Dwork et al., 2006), Federated Learning (FL) (McMahan et al., 2017), and cryptographic methods (Yao, 1982), each playing a distinct conceptual role. Differential Privacy offers a mathematically rigorous way to safeguard information by injecting carefully calibrated noise into data analyses or model training processes. In essence, DP limits the influence of any single data point on the output, so that an attacker cannot reliably infer whether a particular record (for example, one borehole's data in a soil database) was included in the input. This provides a quantifiable privacy guarantee – often expressed via a "privacy budget" ϵ — that balances confidentiality against model accuracy. Federated Learning, in contrast, keeps the raw data in situ at distributed sources and only exchanges learned model parameters or updates. In geotechnical practice, data are naturally partitioned by project site, contractor, or region, reflecting both geological heterogeneity and contractual boundaries. These structural characteristics provide a domain-specific foundation for designing federated learning architectures that respect data ownership while enabling collaborative model development. In a federated learning scenario, multiple parties (e.g. different companies or project sites) collaboratively train a shared model without ever sending their local geotechnical datasets to a central location. By transmitting only aggregated model updates rather than sensitive raw data, FL inherently reduces the risk of exposing proprietary site information and helps meet data protection obligations. Cryptographic approaches such as homomorphic encryption provide yet another layer: they enable computations to be performed directly on encrypted data. This means that even if geotechnical data are outsourced to a third-party cloud or combined in a multi-party analysis, the data remain encrypted throughout the process – the analysts or computing servers never see the plaintext values. Such techniques, while computationally intensive, ensure that confidential information (e.g. exact soil properties or locations) is mathematically shielded from disclosure.

Simply trying out these privacy techniques in a piecemeal or ad-hoc fashion — without a rigorous framework for evaluation — does not constitute a strong academic contribution. A patchwork approach can lead to unclear or unquantified privacy outcomes, offering little confidence to practitioners or regulators. In fact, the absence of standard metrics and methodologies for privacy has been noted to impede effective use of even mature privacy technologies. Therefore, we advocate for establishing a principled standard for evaluating and reporting privacy guarantees in geotechnical ML research. Differential Privacy is particularly well-suited as the foundation for such a standard, given its formal, quantifiable guarantees and its growing recognition as a "gold standard" in data privacy research. Adopting a DP-based framework (for example, reporting the privacy budget ϵ alongside model performance) would enable researchers to rigorously quantify how well sensitive data are protected and to make fair, apples-to-apples comparisons of privacy levels across studies. Murakami et al. (2025) have conducted the first preliminary study that can serve as a demo for future geotechnical research in this direction.

3.2. *Model applicability and engineering practice challenges*

Generalization power and explanability of machine learning models are of major concern to practical engineers when using AI technologies. Two concepts emerge to address this issue. The first idea is combining physical constraints with data-driven models to improve the physical rationality of machine learning models. The second idea is the use of natural language models to provide human-friendly model interpretability and engineering decision support. LLMs have become a promising tool to bridge the communication gap between human engineers and purely quantitative models. Details of these topics are discussed in the following two subsections.

3.2.1. Convergence of PINNs

Deep neural networks (DNNs) have demonstrated strong predictive capabilities across various fields. However, their effectiveness is often constrained by the need for large amounts of high-quality data, increasing the risk of overfitting, particularly in data-scarce scenarios. Such issue can be alleviated by incorporating prior knowledge from known physics. PINNs is one of the possible methods to integrate physics into DNNs, which have emerged as a powerful architecture and have garnered significant attention in recent years (Yuan et al., 2025). PINNs incorporate governing physical laws, typically expressed as ordinary differential equations (ODEs) or partial differential equations (PDEs), directly into the neural network training (Raissi et al., 2019). This framework enables PINNs to effectively solve both forward problems – predicting system states by solving governing equations with known physics and initial/boundary conditions, and inverse problems – identifying unknown parameters from sparse data while maintaining consistency with fundamental principles. By integrating physical constraints, PINNs enhance model interpretability, reduce reliance on extensive datasets, and improve generalization performance.

It is important to clarify that PINNs are not intended to replace established numerical simulation techniques such as finite element or finite difference methods in well-posed forward problems. For deterministic analyses with clearly defined boundary conditions and constitutive models, classical numerical solvers remain the reference standard due to their theoretical maturity and well-established verification frameworks. The primary value of PINNs in geotechnical engineering arises in scenarios where physics and data must be simultaneously assimilated — such as inverse parameter identification from sparse monitoring data, partially observed systems, real-time model updating in digital twin frameworks, or situations requiring rapid surrogate evaluation across varying conditions. In these contexts, PINNs can serve either as hybrid data-physics solvers or as physics-consistent surrogate models, enabling unified optimization that integrates governing equations and measurement data within a single computational framework — an approach that differs fundamentally from traditional simulation followed by separate back-analysis procedures.

The training of PINNs is carried out by minimizing a loss function that enforces both data consistency and physical constraints, typically expressed as a weighted sum

$$L_{\text{total}} = \lambda_{\text{data}} L_{\text{data}} + \lambda_{\text{BC/IC}} L_{\text{BC/IC}} + \lambda_{\text{physics}} L_{\text{physics}}, \tag{3}$$

where L_{data} quantifies the error between network predictions and available data points (if any), $L_{\text{BC/IC}}$ enforces boundary and initial conditions, L_{physics} evaluates the residuals of governing equations at each collocation point within the physical domain, λ_i are weighting factors that control the relative importance of each loss term.

The performance of PINNs is significantly influenced by hyperparameters such as the number and distribution of collocation points as well as the selection of loss function weights λ_i (Wang et al., 2023). An improper selection of weights can lead to imbalanced optimization among different loss terms, causing slow convergence or training instability. As an example, this study investigates how different values of λ_i impact PINN solutions. To demonstrate this, PINNs are employed to estimate the longitudinal behavior of a tunnel subjected to external loading, modeled by a fourth-order ODE. In this work, we specifically focus on the forward problem, meaning that L_{data} in Eq. (3) is omitted. We fix $\lambda_{\text{BC}} = 1$ and vary λ_{ODE} to evaluate its impact on solution accuracy and convergence, as shown in Fig. 6. When $\lambda_{\text{ODE}} = 1$, the network fails to capture the reference deformation obtained with the finite difference method (FDM), as it prioritizes boundary conditions over the governing equation. Increasing λ_{ODE} to 10^9 improves the solution by balancing the loss terms. However, an excessively large value ($\lambda_{\text{ODE}} = 10^{12}$) leads to instability and poor agreement with the FDM, indicating that overemphasizing the physics term can hinder boundary condition satisfaction. These effects are also reflected in the training dynamics. Using the L-BFGS-B optimizer with an adaptive stopping criterion, we observe that small λ_{ODE} values yield fast but suboptimal convergence, while appropriate weighting ensures stable training and accurate predictions. In contrast, very large λ_{ODE} causes fluctuations in the loss, signaling instability.

These results highlight the crucial role of the weight selection for stable and efficient training. Since the training is inherently a trial-and-error process, engineering judgment can be used to assess whether the results are physically reasonable and, if not, weights can be adjusted for retraining. Another strategy is to observe convergence behavior, as a flat or too fluctuant loss curve (see Fig. 6) may indicate poor weight selection or undertraining due to imbalanced constraints. To further enhance performance, self-adaptive weighting strategies such as Neural Tangent Kernel based gradient descent algorithm (Wang et al., 2022) and multi-task learning approaches (Xu et al., 2023) can be employed to dynamically adjust weights during training, ensuring the stability and accuracy in the learning process.

3.2.2. Use of LLMs

The current trend in applying LLMs within geotechnical engineering has predominantly focused on their text generation capabilities, such as text summary (Kumar, 2024), text-based information extraction (Chai et al., 2025), programming code generation (Kim et al., 2024), etc (Wu et al., 2024). Although these applications demonstrate clear immediate benefits, particularly in enhancing operational efficiency and reducing manual efforts on knowledge extraction, they represent merely the initial exploration into LLM's broader capabilities. They essentially treat the LLM as a sophisticated text output tool, rather than a deeply integrated problem-solving partner. In other words, current trends largely reduce LLMs to intelligent scribes for human engineers, tapping only into surface-level capabilities.

Recent advances, however, indicate a transition from text-generation tools toward agentic AI systems, where LLMs are endowed with planning, memory, tool-use, and multi-step reasoning capabilities (Wu et al., 2025). These text-generation applications — useful as they are — do not yet demonstrate possible exploitation of the richer reasoning, planning, and adaptive learning capacities that modern LLM agents can offer. Moving beyond these preliminary uses requires conceptualizing LLMs as autonomous agents embedded within multi-agent systems designed to handle complex geotechnical workflows. Instead of operating in isolation to produce text on demand, an LLM can be situated as a cognitive agent that plans actions, calls tools, and collaborates with other agents (human or AI) to tackle multifaceted engineering problems.

Crucially, language models endowed with agentic capabilities are able to break a high-level task into sub-tasks, dynamically execute each step (e.g. querying data, running analyses, generating code), and iteratively refine outputs – much like an experienced engineer orchestrating a solution. Geotechnical domain knowledge can be explicitly embedded into such agentic systems through structured prompt templates, domain-specific ontologies, rule-based safety constraints, and tool-grounded reasoning workflows that reflect established engineering procedures. Designing these knowledge-augmented agents requires researchers to formalize tacit engineering judgment into computationally interpretable forms, thereby influencing not only application but also the architecture of LLM-based systems. Equally important is the prospect of multi-agent systems: networks of specialized LLM-based agents that communicate in natural language to coordinate their efforts. There have been a few such examples in geotechnical engineering (Li & Shi, 2025; Pang et al., 2025; Xu et al., 2025a,b), but more effort shall be put in this research direction.

To fully realize this vision, rigorous experimental methodologies are essential. Many existing LLM-driven studies lack clearly defined performance metrics and comparative benchmarks, limiting the assessment of true effectiveness and reliability. Thus, future research must prioritize meticulous experimental designs and establish explicit performance criteria for meaningful comparative analyses. For example, we recommend research in this direction to begin adopting some simple performance evaluation practices. In a typical LLM application, prompt engineering

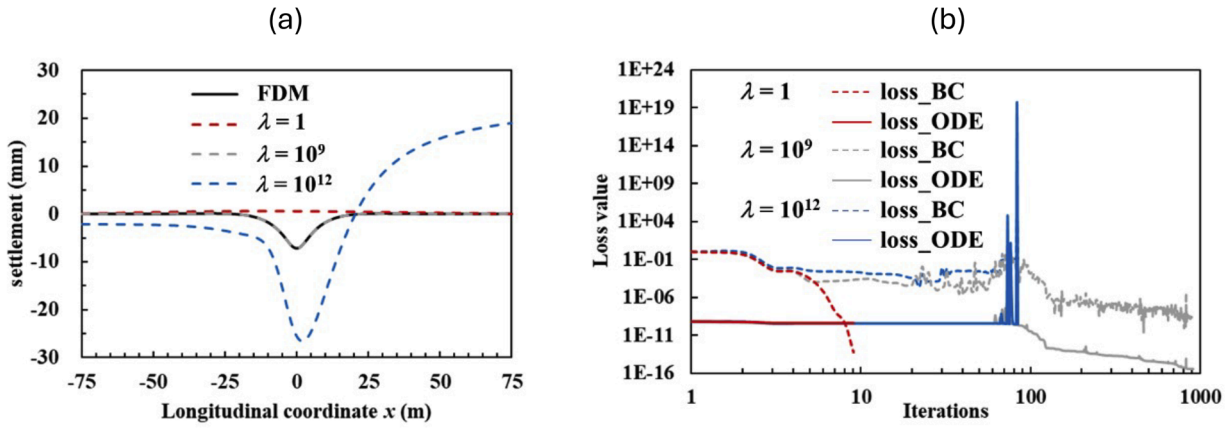


Fig. 6. Influence of the weight λ_{ODE} of the physics-based loss function on (a) the PINNs solution accuracy and (b) the convergence behavior of the loss terms λ_{BC} and λ_{ODE}.

is performed to improve the model performance. This step is analogous to parameter optimization for a machine learning model. Therefore, different datasets should be used for prompt engineering and final evaluation of the designed LLM agent. Additionally, while developing novel agentic AI systems is a critical direction, constructing representative and unbiased benchmark datasets is equally vital. Such datasets will not only facilitate standardized performance evaluations but also support reproducibility and fairness in comparisons across studies. Efforts toward these datasets represent a fundamental contribution to geotechnical AI research, enabling a clearer exploration of AI capabilities and more rigorous validation of LLM-based solutions. Ultimately, the strategic integration of autonomous agent systems, supported by robust experimental frameworks and standardized benchmarks, promises transformative breakthroughs in geotechnical engineering.

4. Future development trends and prospects

As geotechnical engineering continues to embrace AI-driven advancements, several forward-looking trends and strategic directions emerge, emphasizing collaborative research efforts and the integration of interdisciplinary knowledge. In Phoon and Zhang (2023), the authors advocate a data-centric geotechnics perspective that centers geotechnical AI research and application on the nature of geotechnical data. Within this realm, the recently established Future of Machine Learning in Geotechnics (FOMLIG) Council and the associated FOMLIG Workshop series are expected to play a key role in promoting data-centric geotechnics research, practice and regulation. The proactive role of geotechnical researchers, embracing interdisciplinary integration, community-driven open datasets, and the symbiotic fusion of domain expertise with AI methodologies, will define the transformative future of geotechnical engineering. In the following subsections, we elaborate on these key aspects for future developments that are vital in addressing open challenges in geotechnical AI research.

4.1. Building an open foundation through community efforts

Geotechnical researchers should serve as key motivators in developing open foundations for AI integration to benefit the entire research community. Currently, the absence of widely accepted benchmark datasets significantly hinders transparent, reproducible, and comparative studies. It is crucial to foster community-driven initiatives aimed at constructing open-access and high-quality benchmark datasets specifically tailored to geotechnical engineering. Such datasets will provide standard references that enable consistent performance evaluations, promote methodological transparency, and accelerate innovation. Through collective efforts, researchers can mitigate isolated data silos, ensuring more robust, inclusive, and generalized AI applications across the broader community.

A notable example of such community-driven initiatives is the recent emergence of GeoTechathon (Wu et al., 2025)—a series of international hackathon events dedicated to geotechnical engineering. GeoTechathon events gather international teams of geotechnical researchers and engineers to collaboratively explore the potential use of cutting-edge technologies, particularly LLMs and agentic AI systems, within geotechnical contexts. These hackathons provide unique platforms for participants to rapidly prototype innovative solutions, exchange interdisciplinary knowledge, and cultivate collaborative relationships across traditional institutional and geographic boundaries. Importantly, GeoTechathon events encourage the creation of shared resources, openly accessible code, and publicly documented outcomes, promoting transparency and fostering a cooperative environment for future research developments. Such events exemplify how proactive researcher-led initiatives can accelerate the responsible and creative adoption of AI technologies within the geotechnical engineering community.

By continuing to support and replicate similar collaborative and open initiatives, researchers can effectively democratize AI advancements, facilitating rapid technological adoption and innovation throughout the broader geotechnical engineering community.

4.2. Embracing true interdisciplinary research

A pivotal shift from multidisciplinary to truly interdisciplinary research is necessary. Multidisciplinary research often remains compartmentalized, limiting researchers within their established areas of expertise. Interdisciplinary research, however, actively promotes mutual integration of ideas, methodologies, and tools across distinct disciplines. Geotechnical researchers must actively step outside their traditional comfort zones and collaborate meaningfully with experts from other fields such as computer science, applied mathematics, environmental science, and structural engineering. Such integration will ensure the effective translation of emerging AI methods into practical geotechnical solutions and foster novel insights previously inaccessible through isolated disciplinary approaches.

4.3. Fusion of AI capabilities with geotechnical domain knowledge

A critical future direction is the strategic integration of general AI capabilities with geotechnical domain expertise. This fusion will drive innovative breakthroughs in both theory and practice. Particularly promising is the combination of traditional geotechnical mechanics theories with modern AI architectures such as PINNs. These hybrid models can effectively capture the complex, nonlinear behavior inherent in

geotechnical systems, enhancing prediction reliability, interpretability, and overall performance.

Further, distributed collaborative intelligence systems represent a compelling application of AI in managing complex engineering projects, particularly for large-scale geotechnical infrastructure. By leveraging the coordination capabilities of AI, these systems can significantly improve project management efficiency, risk mitigation, and resource allocation across multiple stakeholders and geographically dispersed sites.

Full lifecycle knowledge accumulation enabled by AI technologies will facilitate the continuous refinement of intelligent decision-making systems. Over time, these systems will become increasingly adept at storing, retrieving, and applying historical knowledge, ensuring valuable insights are preserved and leveraged for future projects. Ultimately, the ideal model for future development in geotechnical engineering involves robust human-machine collaboration. Intelligent decision-support frameworks that combine human professional judgment and experience with AI's analytical precision offer powerful solutions for addressing the inherently uncertain and complex nature of geotechnical engineering problems. Such collaborative models promise to reshape decision-making processes and set new benchmarks for industry-wide best practices.

4.4. Proactive role of researchers to strengthen academia-industry collaboration

While methodological innovation is essential, the transformative impact of AI in geotechnical engineering can only be fully realized through validation in real-world projects. Unlike controlled academic environments, practical geotechnical projects operate under contractual liability, budget limitations, construction scheduling pressures, regulatory requirements, and multi-stakeholder risk-sharing frameworks. These constraints fundamentally shape how AI solutions can be deployed.

In this context, geotechnical researchers must take a proactive role in facilitating meaningful academia-industry collaboration. Rather than designing AI research solely around methodological novelty or benchmark performance, researchers should explicitly consider the practical constraints, decision-making workflows, and risk management requirements faced by industry practitioners. This includes:

1. Designing research problems that reflect realistic project scenarios,
2. Providing transparent uncertainty quantification aligned with engineering reliability standards,
3. Developing interpretable models that support accountable decision-making,
4. Structuring datasets and experimental protocols in ways that are compatible with industry data governance and confidentiality constraints.

By proactively incorporating these considerations during research design, geotechnical researchers can lower the barriers for industry adoption and ensure that AI advancements move beyond proof-of-concept demonstrations toward measurable improvements in productivity, safety, and lifecycle performance. In this sense, fostering effective academia-industry collaboration is not merely an external partnership issue, but an intrinsic responsibility of researchers in shaping the future of AI-driven geotechnical engineering.

5. Concluding remarks

This paper has explored the evolving role and emerging responsibilities of geotechnical researchers amid rapid AI advancements. AI technologies significantly enhance information acquisition across investigation, design, construction, and operation phases, leveraging intelligent data-driven models and sensor networks to improve efficiency, accuracy, and reliability in geotechnical practice. However, critical challenges related to data quality, accessibility, privacy, and model applicability and interpretability remain.

Five topics were highlighted in this paper as examples of the opportunities for geotechnical researchers to make contributions. (1) Effective use of transfer learning requires careful validation against risks such as catastrophic forgetting and negative transfer. (2) Probabilistic machine learning models provide robust uncertainty quantification essential for informed decision-making. (3) Addressing data privacy concerns through technologies like Differential Privacy (DP) and Federated Learning (FL) is imperative for secure and collaborative data utilization. (4) PINNs effectively integrate geotechnical domain knowledge, reducing reliance on extensive datasets. (5) LLMs represent a transformative potential beyond simple information extraction, serving as autonomous agents to orchestrate complex workflows and foster richer human-AI collaboration.

Finally, advancing geotechnical AI demands interdisciplinary collaboration and community-driven efforts, including the establishment of shared benchmark datasets and robust validation frameworks. Events like GeoTechathon illustrate successful models for fostering open innovation and collective problem-solving.

In conclusion, by strategically integrating geotechnical domain expertise with AI methodologies, embracing interdisciplinary research, and addressing critical challenges of data management and model validation, geotechnical researchers will play a pivotal role in fully realizing the transformative potential of AI in geotechnical engineering.

CRediT authorship contribution statement

Stephen Wu: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization; **Iason Papaioannou:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization; **Yelu Zhou:** Writing – review & editing, Formal analysis, Data curation; **Yanzhe Liu:** Writing – review & editing, Formal analysis, Data curation; **Mingliang Zhou:** Writing – review & editing, Methodology; **Dongming Zhang:** Writing – review & editing, Writing – original draft, Methodology; **Hongwei Huang:** Writing – review & editing, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Stephen Wu reports financial support was provided by the "Strategic Research Projects" grant from ROIS (Research Organization of Information and Systems). Stephen Wu reports a relationship with The Institute of Statistical Mathematics that includes: employment. Stephen Wu, Iason Papaioannou and Dongming Zhang also holds an editorial role with the journal to which this manuscript has been submitted, which is considered a potential competing interest. All other authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was supported by "Strategic Research Projects" grant from ROIS (Research Organization of Information and Systems) and the National Natural Science Foundation of China (Grant Nos. 52130805). The authors would like to thank the members of the TC304 Committee on Engineering Practice of Risk Assessment and Management of the International Society of Soil Mechanics and Geotechnical Engineering for developing the database 304dB used in this study and making it available for scientific inquiry. We also wish to thank M D'Ignazio and TT Länsivaara for contributing the F-CLAY/7/216 dataset to the TC304 compendium of databases.

References

Azmoon, B., Biniyaz, A., Liu, Z., & Sun, Y. (2021). Image-data-driven slope stability analysis for preventing landslides using deep learning. *IEEE Access*, *9*, 150623–150636.

Baecher, G. B., & Christian, J. T. (2008). Spatial variability and geotechnical reliability. In *Reliability-based design in geotechnical engineering* (pp. 88–145). CRC Press.

Chai, S. H., Chen, I., Huang, J., & Yacoub, T. (2025). Large language model for geotechnical engineering applications using retrieval augmented generation. In *Geotechnical frontiers 2025* , pp. 1–10). https://doi.org/10.1061/9780784485989.001

Chang, J., Huang, H.-w., Thewes, M., Zhang, D.-m., & Wu, H.-m. (2024). Data-based postural prediction of shield tunneling via machine learning with physical information. *Computers and Geotechnics*, *174*, 106584. https://doi.org/10.1016/j.compgeo.2024.106584

Cheng, X., Liu, X., Qiao, H.-H., Wu, Z.-Q., & Cai, G. J. (2024). A new SPT-CPT correlation by statistical analysis based on a database from Anhui, China. *Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards*, *18*, 838–853. https://doi.org/10.1080/17499518.2023.2280456

De Lange, M., Aljundi, R., Masana, M., Parisot, S., Jia, X., Leonardis, A., Slabaugh, G., & Tuytelaars, T. (2022). A continual learning survey: Defying forgetting in classification tasks. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *44*(7), 3366–3385. https://doi.org/10.1109/TPAMI.2021.3057446

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. In *International conference on learning representations (ICLR)*.

Dwork, C., McSherry, F., Nissim, K., & Smith, A. (2006). Calibrating noise to sensitivity in private data analysis. In *Theory of cryptography conference* (pp. 265–284). Springer. https://doi.org/10.1007/11681878_14

D'Ignazio, M., Phoon, K.-K., Tan, S. A., & Länsivaara, T. T. (2016). Correlations for undrained shear strength of Finnish soft clays. *Canadian Geotechnical Journal*, *53*(10), 1628–1645.

Gressmann, F., Király, F. J., Mateen, B., & Oberhauser, H. (2018). Probabilistic supervised learning. *CoRR*, *abs/1801.00753*. https://arxiv.org/abs/1801.00753.

Griffiths, D. V., Huang, J., & Fenton, G. A. (2009). On the reliability of earth slopes in three dimensions. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, *465*(2110), 3145–3164.

Huang, H., Liu, S., Zhou, M., Shao, H., Li, Q., & Thansirichaisree, P. (2025). Automated 3D defect inspection in shield tunnel linings through integration of image and point cloud data. *AI in Civil Engineering*, *4*, 12. https://doi.org/10.1007/s44150-025-00012-3

Huang, H.-w., Wang, C.-s., Zhou, M.-l., & Qu, L.-q. (2024). Compressive strength detection of tunnel lining using hyperspectral images and machine learning. *Tunnelling and Underground Space Technology*, *153*. https://doi.org/10.1016/j.tust.2024.105743

Jiang, Q.-H., Zhang, J.-Z., Zhang, D.-M., & Huang, H. W. (2024). Simulation of geological uncertainty based on improved three-dimensional coupled Markov chain model. *Engineering Geology*, *340*, 107647. https://doi.org/10.1016/j.enggeo.2024.107647

Kim, D., Kim, T., Kim, Y., Byun, Y.-H., & Yun, T. S. (2024). A ChatGPT-MATLAB framework for numerical modeling in geotechnical engineering applications. *Computers and Geotechnics*, *169*, 106237. https://doi.org/10.1016/j.compgeo.2024.106237

Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwińska, A. et al. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, *114*(13), 3521–3526. https://doi.org/10.1073/pnas.1611835114

Kumar, K. (2024). Geotechnical parrot tales (GPT): Harnessing large language models in geotechnical engineering. *Journal of Geotechnical and Geoenvironmental Engineering*, *150*(1), 02523001. https://doi.org/10.1061/JGGEFK.GTENG-11828

Lei, Y., Huang, J., Cui, Y., Jiang, S.-H., Wu, S., & Ching, J. (2023). Time capsule for landslide risk assessment. *Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards*, *17*(4), 613–634.

Li, H., & Shi, C. (2025). Few-shot learning of geological cross-sections from sparse data using large language model. *Geodata and AI*, *2*, 100010. https://doi.org/10.1016/j.geoai.2025.100010

McCloskey, M., & Cohen, N. J. (1989). Catastrophic interference in connectionist networks: The sequential learning problem. In *Psychology of learning and motivation* (pp. 109–165). Academic Press (*vol. 24*). https://doi.org/10.1016/S0079-7421(08)60536-8

McMahan, B., Moore, E., Ramage, D., Hampson, S., & y Arcas, B. A. (2017). Communication-efficient learning of deep networks from decentralized data. In *Artificial intelligence and statistics* (pp. 1273–1282). PMLR.

Mikami, H., Takenaka, S., Ueda, Y., Matsui, Y., Yamasaki, T., & Aizawa, K. (2022). A scaling law for synthetic-to-real transfer: How much is your pre-training effective? In *International conference on learning representations (ICLR)*. https://openreview.net/forum?id=vpQF0YwFqC.

Murakami, T., Wu, S., Zhang, D.-M., Asano, K., Otake, Y., & Phoon, K. (2025). Differential privacy in geotechnical engineering. *Geodata and AI*, *1*, 100004.

Murphy, K. P. (2012). Machine learning: A probabilistic perspective. MIT Press.

Nie, C., Zhang, D.-M., Ouyang, L.-H., Huang, X., Zhang, B., & Tong, Y. (2023). Dynamic prediction of longitudinal settlement of existing tunnel using convres-DLinear model with integration of undercrossing construction process information. *Geosciences*, *13*. https://doi.org/10.3390/geosciences1308xxx

Nie, L.-C., Song, Z.-C., Li, Z.-Q., & Zhang, S. L. (2024). Ground-tunnel geological prospecting and treatment methods of small-diameter TBM crossing shallow buried water-rich tunnel: A case study. *Tunnelling and Underground Space Technology*, *153*. https://doi.org/10.1016/j.tust.2024.105744

Pang, H., Lo, M. K., Leung, A. Y. F., & Wu, S. (2025). Reconstruction of landslide events using LLM-based agentic AI with multimodal data. *SSRN* 5340542.

Phoon, K.-K., Cao, Z.-J., Ji, J., Leung, Y. F., Najjar, S., Shuku, T., Tang, C., Yin, Z.-Y., Ikumasa, Y., & Ching, J. (2022). Geotechnical uncertainty, modeling, and decision making. *Soils and Foundations*, *62*(5), 101189.

Phoon, K.-K., & Kulhawy, F. H. (1999). Characterization of geotechnical variability. *Canadian Geotechnical Journal*, *36*(4), 612–624.

Phoon, K.-K., & Zhang, W. (2023). Future of machine learning in geotechnics. *Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards*, *17*(1), 7–22.

Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019). Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational physics*, *378*, 686–707.

Wang, S., Sankaran, S., Wang, H., & Perdikaris, P. (2023). An expert's guide to training physics-informed neural networks. arXiv preprint arXiv:2308.08468.

Wang, S., Yu, X., & Perdikaris, P. (2022). When and why PINNs fail to train: A neural tangent kernel perspective. *Journal of Computational Physics*, *449*, 110768.

Wang, Z., Dai, Z., Póczos, B., & Carbonell, J. (2019). Characterizing and avoiding negative transfer. In *Proceedings of the IEEE/CVPR*, 11285–11294.

Wu, S., Otake, Y., Mizutani, D., Liu, C., Asano, K., Sato, N., Saito, T., Baba, H., Fukunaga, Y., Higo, Y., Kamura, A., Kodama, S., Metoki, M., Nakamura, T., Nakazato, Y., Shioi, A., Takenobu, M., Tsukioka, K., & Yoshikawa, R. (2025). Future-proofing geotechnics workflows: Accelerating problem-solving with large language models. *Georisk: Assessment and Management of Risk for Engineered Systems and Geohazards*, *19*(2), 307–324.

Wu, S., Shi, C., Leung, A., Otake, Y., Konishi, C., Zhou, M., Tao, Y., Cao, Z., & Nakamura, T. (2025). Perspectives : LLM agents reshaping the foundation of geotechnical problem-solving. *Geodata & AI*, *4*, 100036.

Xia, Y.-X., Zhang, D.-M., Zhang, B., Huang, H.-W., Tong, Y., & Nie, C. (2025). Real-time prediction of tunnel face deformations by multivariate construction information. *Tunnelling and Underground Space Technology*, *164*. https://doi.org/10.1016/j.tust.2025.105800

Xu, C., Cao, B. T., Yuan, Y., & Meschke, G. (2023). Transfer learning based physics-informed neural networks for solving inverse problems in engineering structures under different loading scenarios. *Computer Methods in Applied Mechanics and Engineering*, *405*, 115852.

Xu, H.-R., Zhang, N., Yin, Z.-Y., & Guy Atangana Njock, P. (2025a). GeoLLM: A specialized large language model framework for intelligent geotechnical design. *Computers and Geotechnics*, *177*, 106849. https://doi.org/10.1016/j.compgeo.2024.106849

Xu, H.-R., Zhang, N., Yin, Z.-Y., & Njock, P. G. A. (2025b). Multimodal framework integrating multiple large language model agents for intelligent geotechnical design. *Automation in Construction*, *176*, 106257. https://doi.org/10.1016/j.autcon.2025.106257

Yao, A. C. (1982). Protocols for secure computations. In *23rd Annual symposium on foundations of computer science (FOCS)* (pp. 160–164). IEEE. https://doi.org/10.1109/SFCS.1982.88

Yeh, C.-H., Lu, Y.-C., Khoshnevisan, S., Juang, C.-H., Tien, Y.-M., & Dong, J. J. (2024). LiDAR-based 3D litho-stratigraphic models calibrated with limited boreholes. *Engineering Geology*, *332*. https://doi.org/10.1016/j.enggeo.2023.107217

Yuan, B., Choo, C. S., Yeo, L. Y., Wang, Y., Yang, Z., Guan, Q., Suryasentana, S., Choo, J., Shen, H., Megia, M. et al. (2025). Physics-informed machine learning in geotechnical engineering: A direction paper. *Geomechanics and Geoengineering*, *20*(5), 1128–1159.

Zhang, D.-M., Fu, L., Huang, H.-W., Wu, H.-M., & Li, G. (2023a). Deep learning-based automatic detection of muck types for earth pressure balance shield tunneling in soft ground. *Computer-Aided Civil and Infrastructure Engineering*, *38*, 940–955. https://doi.org/10.1111/mice.12946

Zhang, D.-M., Nie, C., Chen, M.-Q., Huang, H.-W., & Wu, Y. (2023b). Wireless tilt sensor based monitoring for tunnel longitudinal settlement: Development and application. *Measurement*, *217*. https://doi.org/10.1016/j.measurement.2023.113795

Zhang, D.-M., Nie, C., Zhang, J.-Z., Huang, H.-W., & Huang, X. (2024a). Consortium blockchain-based tunnel data bank for traceable sharing and treatment of structural health monitoring data. *Automation in Construction*, *167*. https://doi.org/10.1016/j.autcon.2024.104243

Zhang, D.-M., Zhang, J.-Z., Huang, H.-W., Qi, C.-C., & Chang, C. Y. (2020). Machine learning-based prediction of soil compression modulus with application of 1D settlement. *Journal of Zhejiang University-Science A*, *21*, 430–444. https://doi.org/10.1631/jzus.A1900245

Zhang, D.-M., Zhou, Y.-L., Huang, H.-W., & Zhang, J. Z. (2023c). A physics and information dual-driven intelligent diagnosis method for longitudinal mechanical behavior of long-distance shield tunnels. *Rock and Soil Mechanics*, *44*, 2997–3010. https://doi.org/10.1000/rocksoilm2023xxx

Zhang, D.-M., Zhu, S.-D., Zhou, M.-L., Huang, H.-W., & Tong, Y. (2024b). Damage quantification and failure prediction of rock: A novel approach based on energy evolution obtained from infrared radiation and acoustic emission. *International Journal of Rock Mechanics and Mining Sciences*, *183*. https://doi.org/10.1016/j.ijrmms.2024.105900

Zhang, J.-Z., Huang, H.-W., Zhang, D.-M., Phoon, K.-K., Liu, Z.-Q., & Tang, C. (2021a). Quantitative evaluation of geological uncertainty and its influence on tunnel structural performance using improved coupled Markov chain. *Acta Geotechnica*, *16*, 3709–3724. https://doi.org/10.1007/s11440-021-01029-6

Zhang, J.-Z., Phoon, K.-K., Zhang, D.-M., Huang, H.-W., & Tang, C. (2021b). Novel approach to estimate vertical scale of fluctuation based on CPT data using convolutional neural networks. *Engineering Geology*, *294*. https://doi.org/10.1016/j.enggeo.2021.106534

Zhang, P., Yin, Z., & Jin, Y. (2022). Machine learning-based modelling of soil properties for geotechnical design: Review, tool development and comparison. *Archives of Computational Methods in Engineering*, *29*, 1229–1245. https://doi.org/10.1007/s11831-021-09615-5

Zhang, S., Ding, L., Xie, M., He, X., Yang, R., & Tong, C. (2024c). Reliability analysis of slope stability by neural network, principal component analysis, and transfer learning techniques. *Journal of Rock Mechanics and Geotechnical Engineering*, *16*. https://doi.org/10.1016/j.jrmge.2023.06.023

Zhou, M.-L., Wang, C.-S., Huang, H.-W., Chen, W., Shao, H., & Zhang, D. M. (2023). Safety state evaluation of operational shield tunnel structures by integrating computer vision and performance analysis. *Journal of Basic Science and Engineering*, *31*, 1461–1476. https://doi.org/10.3969/j.issn.1001-5118.2023.12.007

Zinas, O., Papaioannou, I., Schneider, R., & Cuéllar, P. (2025). Multivariate Gaussian process regression for 3D site characterization from CPT and categorical borehole data. *Engineering Geology*, *352*, 108052.